UF UNIVERSITY of FLORIDA

COP5615 — DISTRIBUTED OPERATING SYSTEMS PRINCIPLES

Project 2

Bonus: Failure Model

Permanent Connection Failures

Pradhyuman Singh Shekhawat

October 2026

1 Objective

The bonus extends the Gossip Simulator with a parameter-controlled permanent connection-failure model. The experiment investigates how randomly removing connections affects network connectivity and the convergence of Gossip and Push-Sum across two topologies.

2 Implementation

Let $p \in [0, 1]$ denote the probability that an undirected connection fails permanently. The failure model first obtains the topology's undirected edges. Each edge is independently retained with probability $1 - p$ or removed with probability p. A removed edge is deleted from both endpoint neighbor lists, ensuring that neither endpoint can use the failed connection.

Before starting the algorithm, the coordinator checks whether the remaining graph is connected. If it is disconnected, the trial is recorded as `disconnected`; otherwise, the selected algorithm runs on the surviving graph. This is a permanent topology change for the duration of a trial, rather than a random decision to drop each individual message.

3 Experimental Setup

The experiments use nine nodes and the `full` and `2D` topologies. Both Gossip and Push-Sum were evaluated at $p \in \{0, 0.1, 0.25, 0.5, 0.75, 1.0\}$, with ten trials per configuration: $2 \times 2 \times 6 \times 10 = 240$ trials in total. Each run used a 5,000 ms convergence timeout.

The CSV file records the status and convergence time of each trial. Successful convergence, disconnection, timeout, and error are treated as distinct outcomes. Mean convergence time is calculated only from successful trials, so it should not be interpreted as the expected completion time across all trials.

4 Results

Table 1 and Figures 1–2 are generated from the recorded CSV results. The plots separate the two topologies into panels to make the Gossip and Push-Sum trends distinguishable.

Algorithm	Topology	p	Success	Disconnected	Timeout	Mean time (μs)
Gossip	full	0	10/10	0/10	0/10	28.9
Gossip	full	0.1	10/10	0/10	0/10	28.5
Gossip	full	0.25	9/10	0/10	1/10	44.4
Gossip	full	0.5	7/10	0/10	3/10	102.3
Gossip	full	0.75	1/10	5/10	4/10	128.0
Gossip	full	1	0/10	10/10	0/10	–
Gossip	2D	0	6/10	0/10	4/10	95.5
Gossip	2D	0.1	3/10	1/10	6/10	95.0
Gossip	2D	0.25	3/10	5/10	2/10	95.7
Gossip	2D	0.5	0/10	10/10	0/10	–
Gossip	2D	0.75	0/10	10/10	0/10	–
Gossip	2D	1	0/10	10/10	0/10	–
Push-Sum	full	0	10/10	0/10	0/10	1825.6
Push-Sum	full	0.1	10/10	0/10	0/10	1271.9
Push-Sum	full	0.25	10/10	0/10	0/10	1431.4
Push-Sum	full	0.5	10/10	0/10	0/10	1642.3
Push-Sum	full	0.75	4/10	6/10	0/10	3497.8
Push-Sum	full	1	0/10	10/10	0/10	–
Push-Sum	2D	0	10/10	0/10	0/10	2737.8
Push-Sum	2D	0.1	9/10	1/10	0/10	2899.3
Push-Sum	2D	0.25	5/10	5/10	0/10	3684.6
Push-Sum	2D	0.5	2/10	8/10	0/10	5182.5
Push-Sum	2D	0.75	0/10	10/10	0/10	–
Push-Sum	2D	1	0/10	10/10	0/10	–

Table 1: Results for ten trials per configuration. Mean convergence time includes successful trials only.

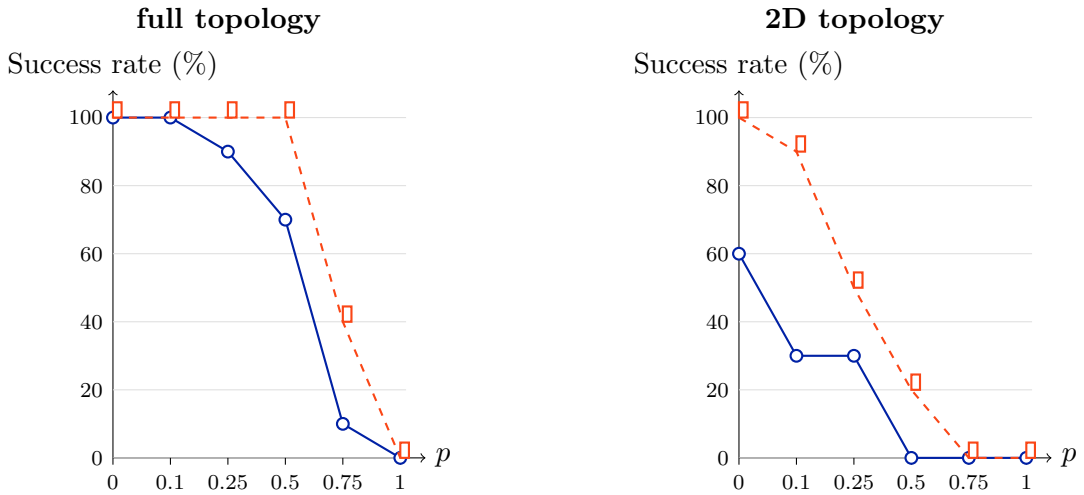


Figure 1: Successful convergence rate as connection-failure probability increases. Each point summarizes ten trials.

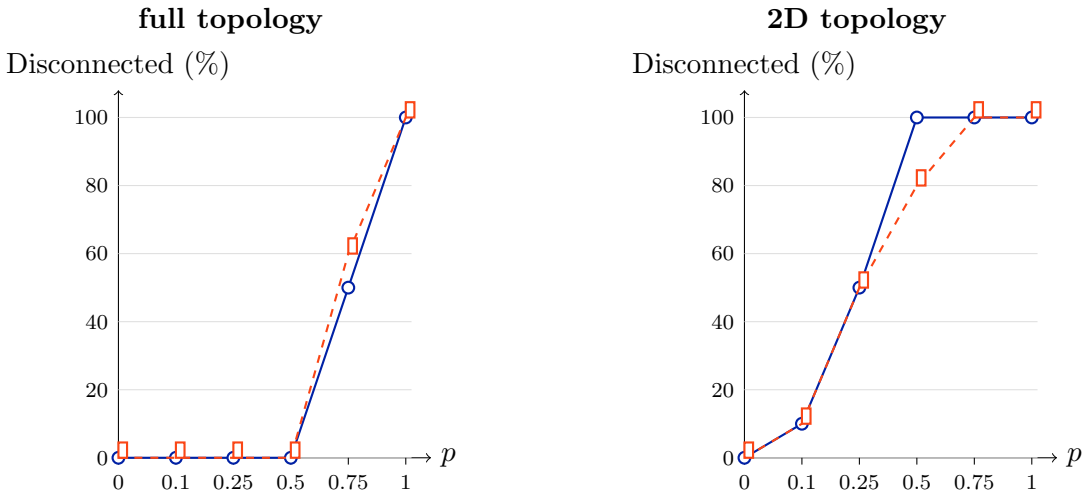


Figure 2: Fraction of trials whose remaining graph was disconnected. Each point summarizes ten trials.

5 Observations

The results show that topology affects tolerance to permanent connection failures. In the full topology, Push-Sum converged in all ten trials at each tested probability from $p = 0$ through $p = 0.5$. At $p = 0.75$, it converged in four of ten trials. In the 2D topology, Push-Sum success decreased from ten of ten trials at $p = 0$, to five of ten at $p = 0.25$, and two of ten at $p = 0.5$.

Gossip also showed a sharper decline in the 2D topology: it converged in three of ten trials at $p = 0.1$ and $p = 0.25$, and none at $p \geq 0.5$. For the full topology, Gossip converged in all ten trials at $p = 0$ and $p = 0.1$, and seven of ten at $p = 0.5$.

These observations are consistent with the full topology having more alternative paths between nodes. Removing edges from a grid can more readily disconnect it or leave a graph on which the algorithm fails to finish before the timeout. The experiment supports this conclusion for the tested graph size, parameter values, and random trials; it does not establish a universal ranking for every network size.

6 Limitations

The experiments use ten random trials per configuration, so measured rates can vary with the sampled failed edges. A timeout means that the algorithm did not report convergence within five seconds; it does not prove that convergence is impossible. A disconnected graph is reported separately because the coordinator rejects it before the algorithm runs. This experiment models permanent connection failures; it does not implement node crashes or temporary connection recovery.

7 Conclusion

The bonus implements a parameter-controlled permanent undirected connection-failure model, tests its effects on Gossip and Push-Sum, and records the outcomes over 240 trials. The observed results show that the full topology generally retains successful convergence at higher tested

failure probabilities than the 2D topology. The conclusion is limited to the tested configurations and trial counts.